TORFRESMA

Cliente	Lacticínios Tirol Ltda
Cidade cliente	Ipiranga - PR
Os	18869 e 18871
Descrição do projeto	Diagrama Pneumático Sistema de Paletização Robotizada

Potência		Tensão de controle	24 VDC
Corrente		Frequência	
Tensão		Nº de série	

Projetista	Eduardo Augusto	Criado em	11/09/2024
Revisão	3.0	Editado em	28/10/2024

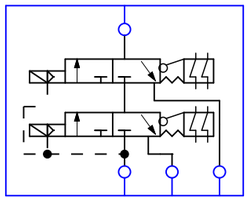
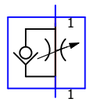
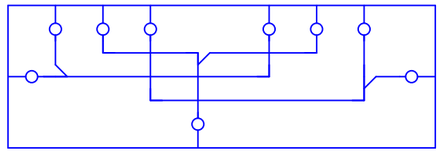
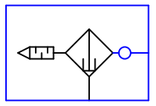
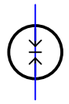
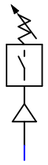
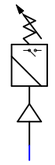
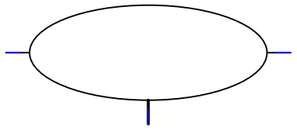
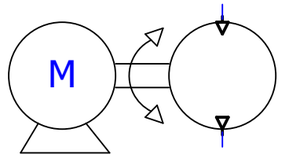
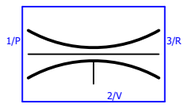
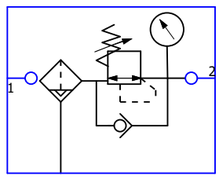
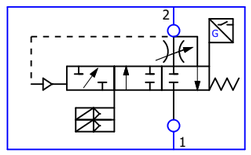
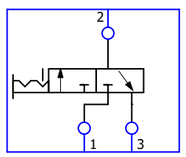
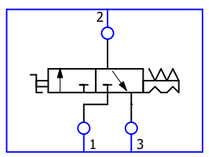
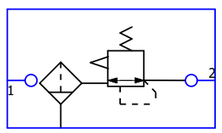
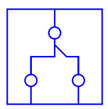
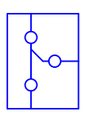
Quantidade de páginas 26

País: Brasil Cep: 89900-000 Rua Arno Erich Schwambach, 910 - Progresso, São Miguel do Oeste - SC Telefone: (49) 3631-4200

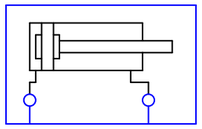
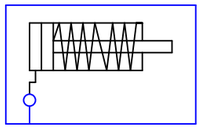
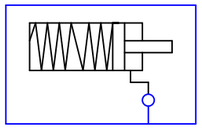
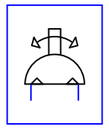
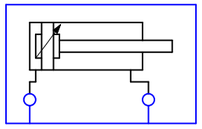
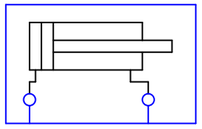
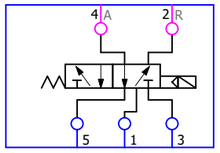
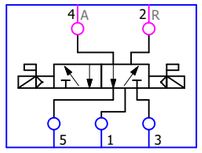
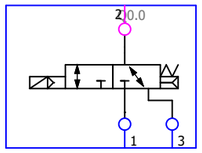
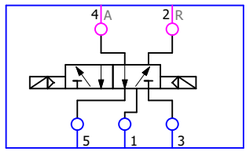
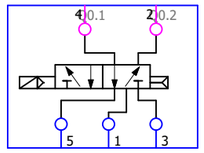
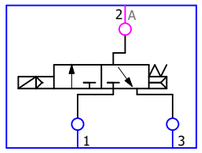
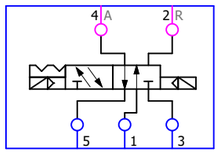
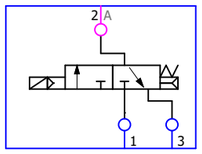
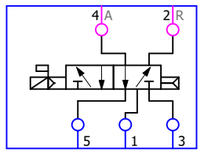
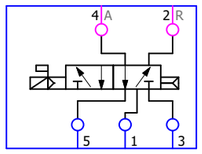
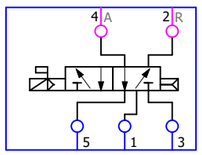
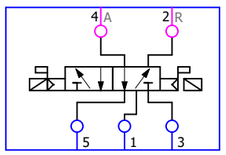
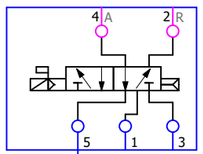
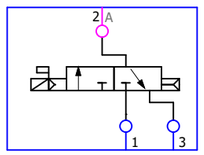
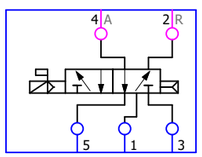
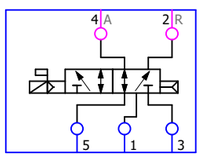
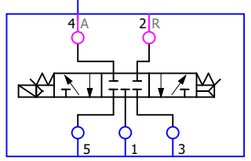
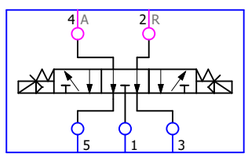
Índice

Estrutura da página	Número da página	Descrição da página
Relatórios iniciais	1	Capa
Relatórios iniciais	2	Índice
Relatórios iniciais	3	Legenda simbologia
Relatórios iniciais	4	Legenda simbologia
Relatórios iniciais	5	Legenda simbologia
Módulo geral	6	Quadro pneumático - QPU1
Módulo geral	7	Reserva
Módulo geral	8	Reserva
Módulo geral	9	Quadro pneumático - QPU9
Módulo geral	10	Quadro pneumático - QPU9
Módulo geral	11	Quadro pneumático - QPU9
Módulo geral	12	Reserva
Módulo geral	13	Quadro pneumático - QPU18
Módulo geral	14	Quadro pneumático - QPU18
Módulo geral	15	Quadro pneumático - QPU18
Módulo geral	16	Reserva
Módulo geral	17	Quadro pneumático - QPU25
Módulo geral	18	Quadro pneumático - QPU25
Módulo geral	19	Quadro pneumático - QPU25
Módulo geral	20	Reserva
Módulo geral	21	Reserva
Módulo geral	22	Reserva
Módulo geral	23	Garra Robô
Módulo geral	24	Garra Robô
Módulo geral	25	Reserva
Módulo geral	26	Reserva

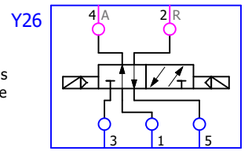
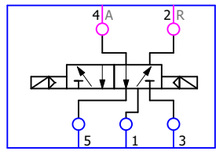
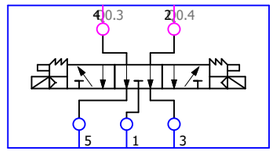
Legenda simbologia

Rede de Ar Cliente	YS1 Válvula dupla de segurança SMC 	Regulador de fluxo 	MFD1 Manifold 
Silenciador com dreno 	Passa muro 	Pressostato B1 	Vacuostato B2 
Válvula de retenção	R1 Reservatório 	Silenciador	Bomba de Vácuo 
Ventosa	Gerador de Vácuo 	Filtro com regulador 	Válvula de Segurança Festo YS2 
V1 Válvula ON/OFF 	V2 Válvula direcional Manual 	Filtro com Pressostato 	Conexão tipo T 
Conexão tipo Y 			

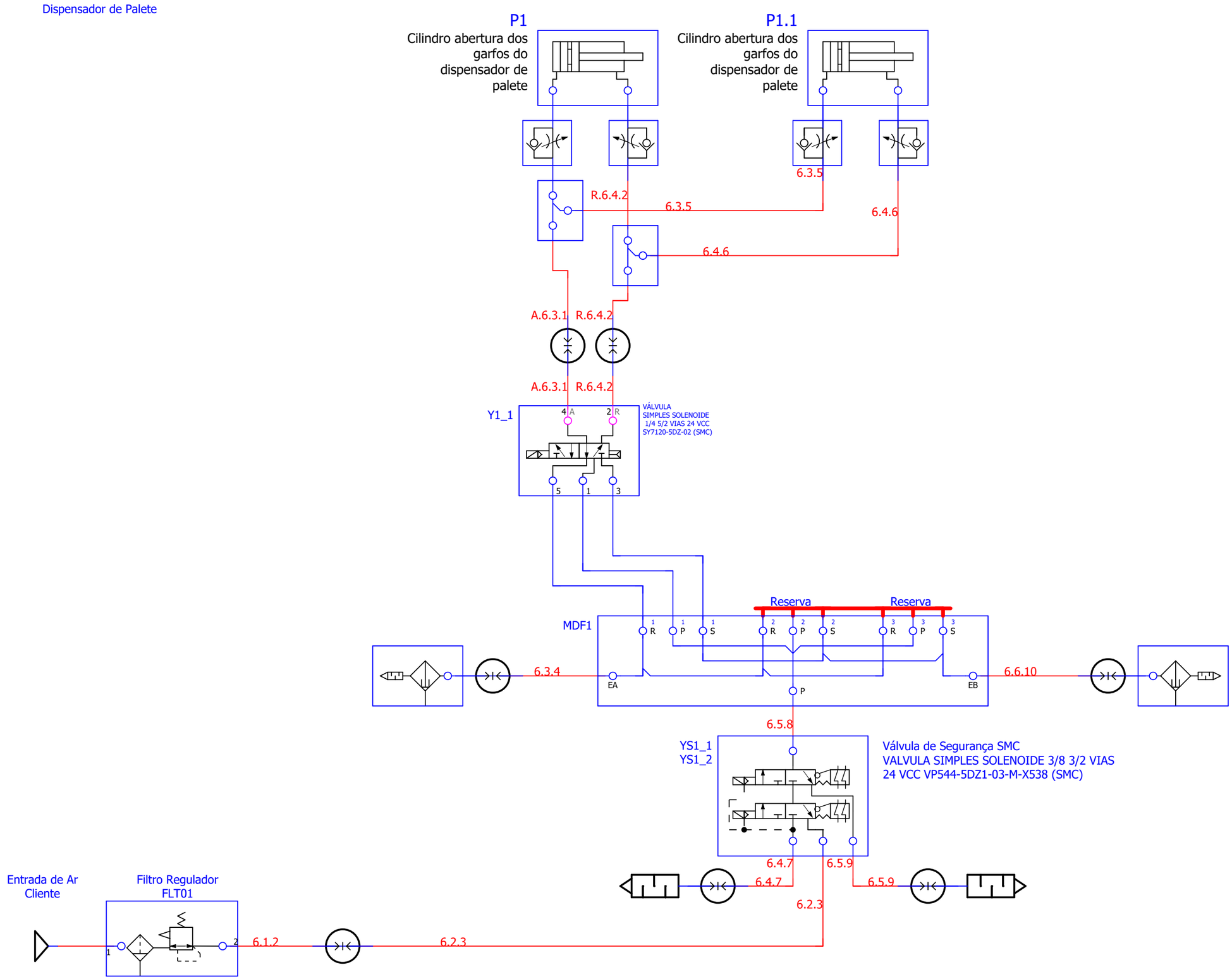
Legenda simbologia

Cilindro dupla ação Avanço e retorno por pressão P1 	Cilindro de simples ação com retorno por mola P2 	Cilindro de simples ação com avanço por mola P3 	Cilindro pneumático rotativo P4 
Cilindro de dupla ação com amortecimento P5 	Cilindro de dupla ação P6 	Valvula 5/2 vias Y2 	Válvula 5/2 dupla solenoide Festo Y3 
Válvula 3/2 vias simples solenoide retorno por mola para vácuo Festo Y4 	Válvula 5/2 vias dupla solenoide SMC Y5 	Válvula 5/2 vias simples solenoide SMC Y6 	Válvula 3/2 vias simples solenoide SMC Y7 
Válvula 5/2 vias dupla solenoide SMC Y12 	Cilindro dupla ação Avanço e retorno por pressão Y13 	Cilindro dupla ação Avanço e retorno por pressão Y16 	Cilindro dupla ação Avanço e retorno por pressão Y18 
Válvula 3/2 vias simples solenoide retorno por mola para vácuo Y20 	Válvula 5/2 vias dupla solenoide Y19 	Válvula 5/2 vias simples solenoide Festo Y17 	Válvula3/2 vias simples solenoide normalmente fechada Festo Y21 
Válvula 5/2 vias simples solenoide Festo Y22 	Válvula5/2 vias simples solenoide Festo Y23 	Válvula 5/3 vias dupla solenoide centro fechado SMC Y8 	Válvula 5/3 vias dupla solenoide centro aberto negativo SMC Y14 

Legenda simbologia

Válvula 5/2 vias dupla solenoide avanço SMC 	Válvula 5/2 vias dupla solenoide recuo SMC 	Válvula 5/3 vias dupla solenoide centro aberto FESTO 	

Dispensador de Palete

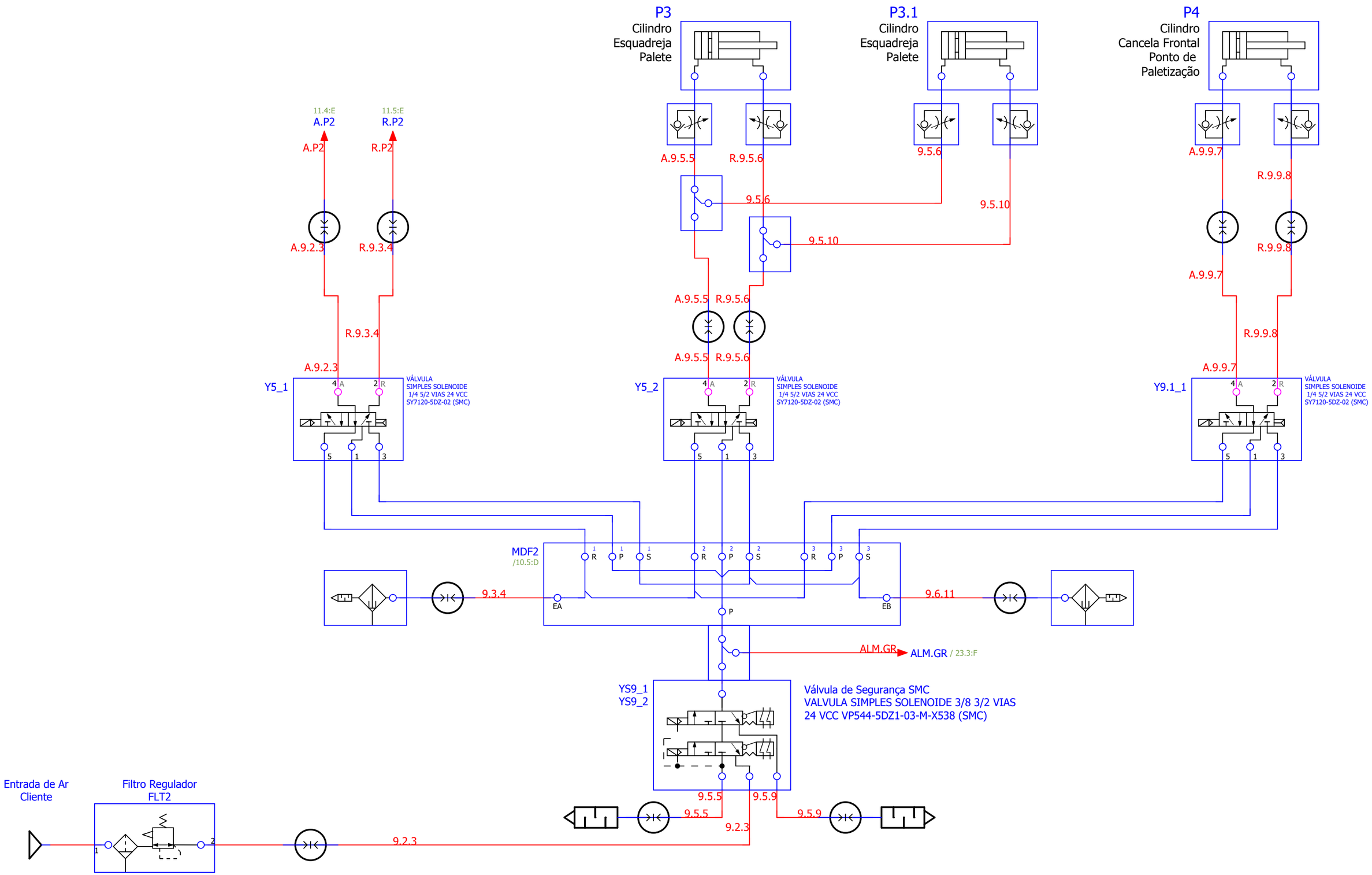


TORFRESMA	Data	28/10/2024	Diagrama Pneumático Sistema de Paletização Robotizada	Quadro pneumático - QPU1	Lacticínios Tirol Ltda	+ Módulo geral
	Editor	Eduardo Augusto			18869 e 18871	Ipiranga - PR	Página 6 / 26
	Versão	3.0				

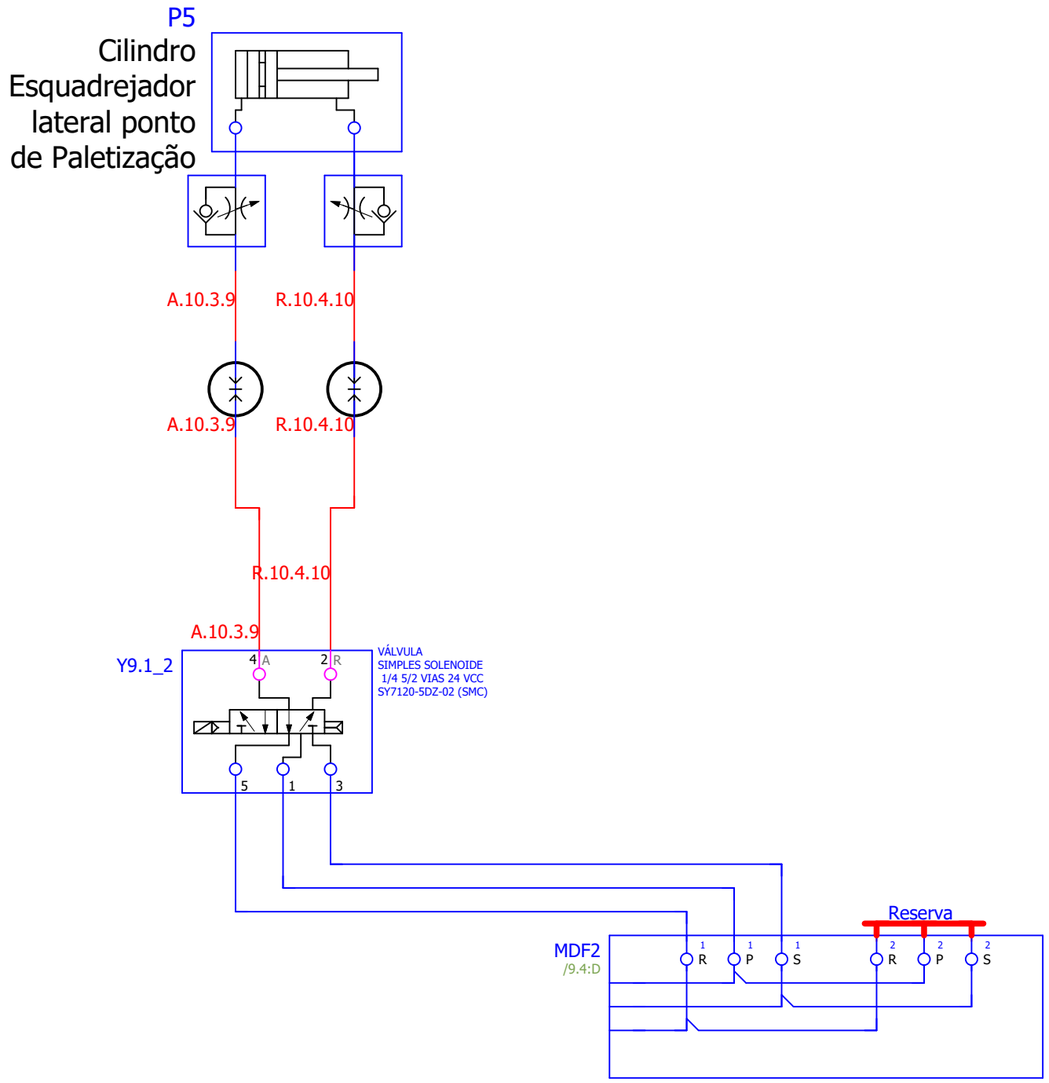
Reserva

Reserva

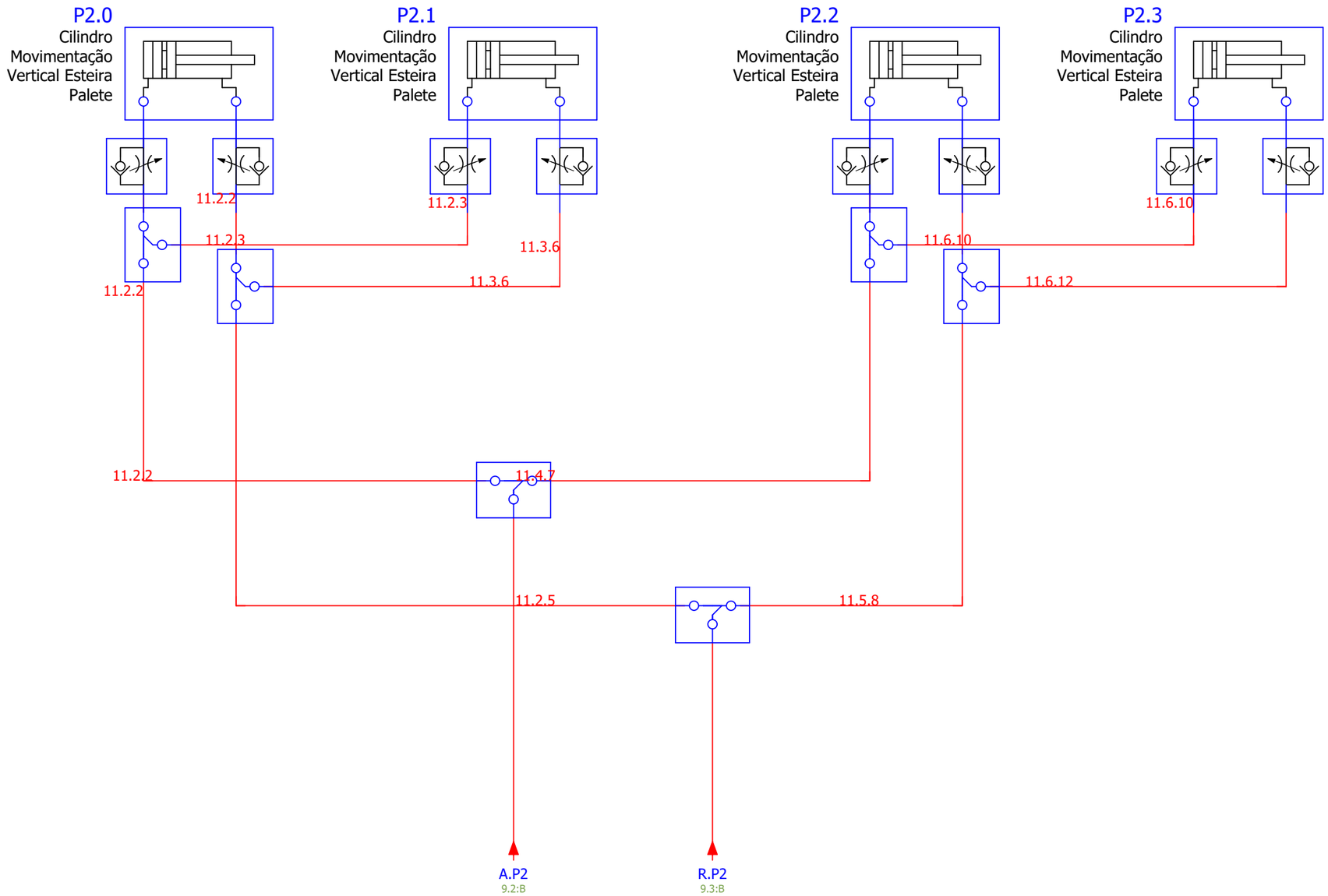
Célula de paletização Trepkus



TORFRESMA	Data	28/10/2024	Diagrama Pneumático Sistema de Paletização Robotizada	Quadro pneumático - QPU9	Lacticínios Tirol Ltda	+ Módulo geral
	Editor	Eduardo Augusto			18869 e 18871	Ipiranga - PR	Página 9 / 26
	Versão	3.0				



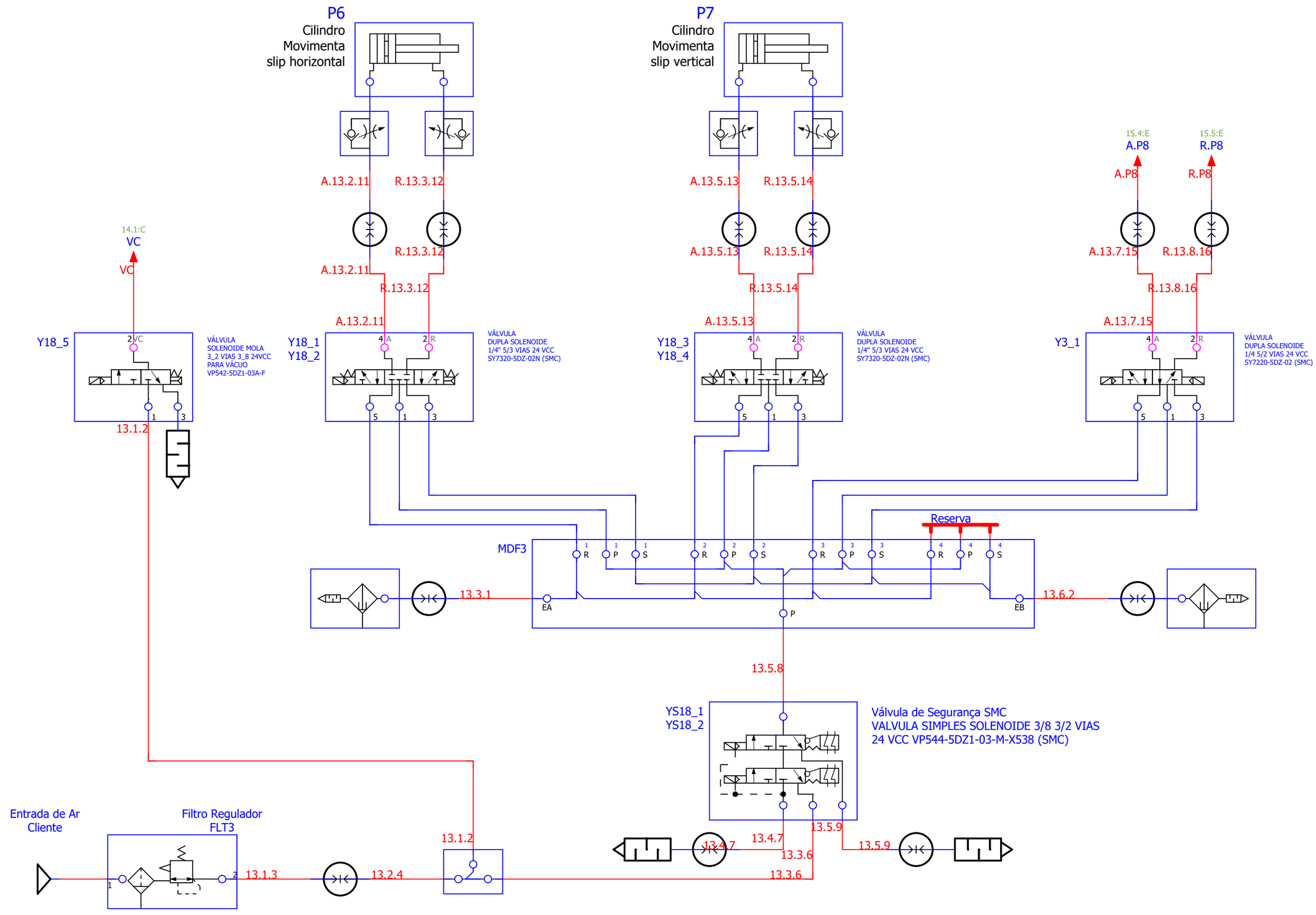
TORFRESMA	Data	28/10/2024	Diagrama Pneumático Sistema de Paletização Robotizada	Quadro pneumático - QPU9	Lacticínios Tirol Ltda	+ Módulo geral
	Editor	Eduardo Augusto			18869 e 18871	Ipiranga - PR	Página 10 / 26
	Versão	3.0				



TORFRESMA	Data	28/10/2024	Diagrama Pneumático Sistema de Paletização Robotizada	Quadro pneumático - QPU9	Lacticínios Tirol Ltda	+ Módulo geral
	Editor	Eduardo Augusto			18869 e 18871 / Ipiranga - PR	Página 11 / 26
	Versão	3.0				

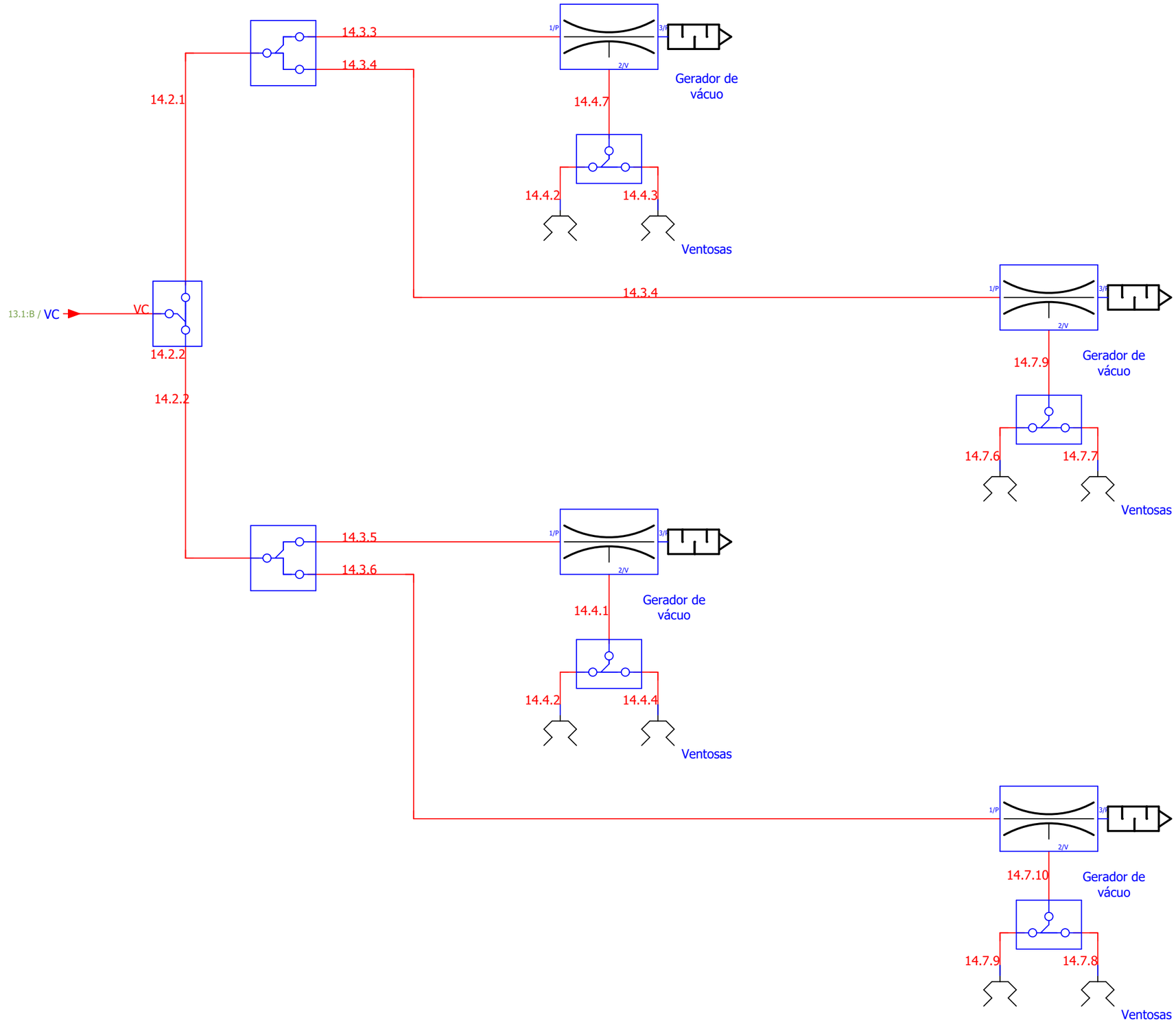
Reserva

Aplicador de papelão

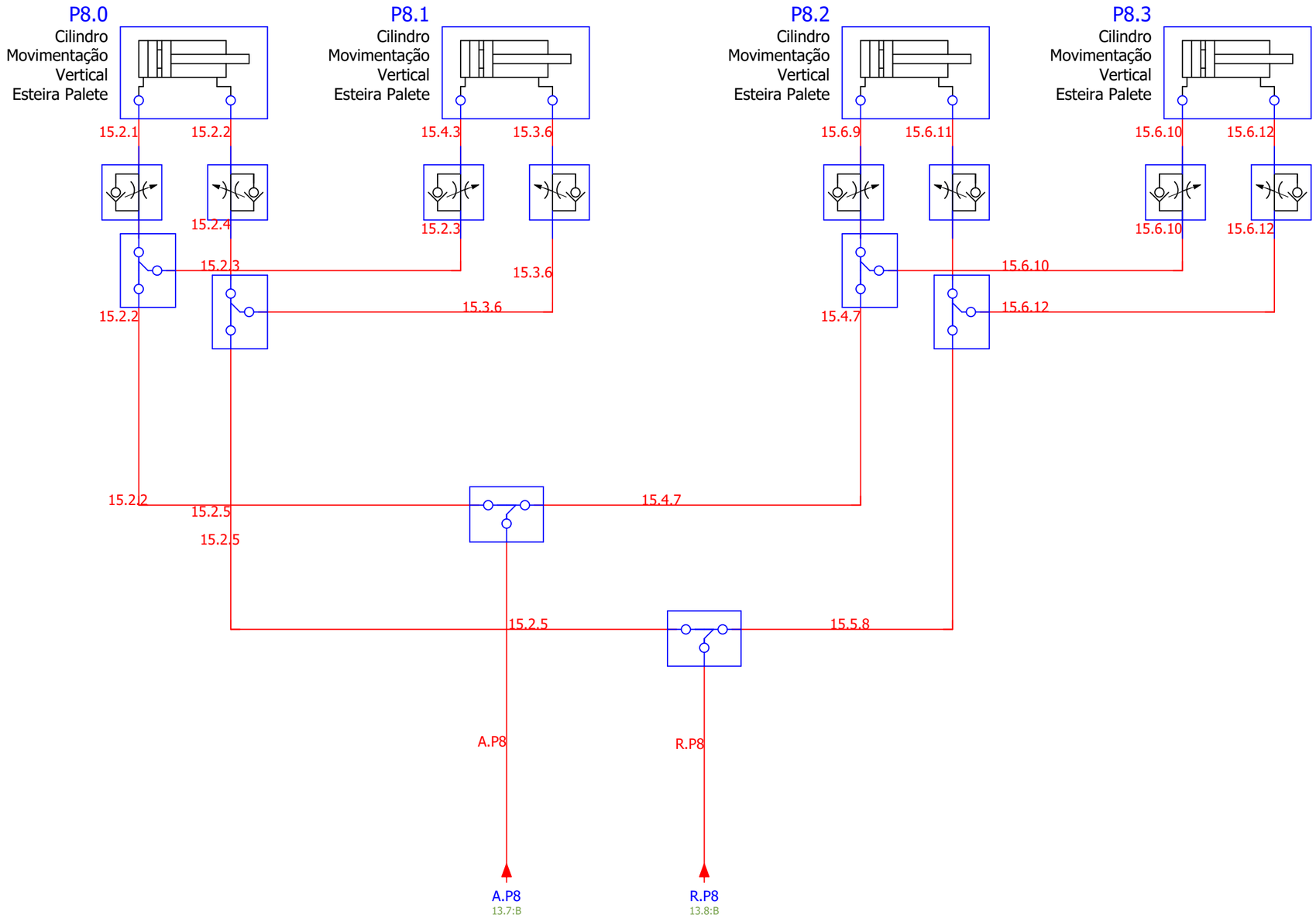


TORFRESMA	Data	28/10/2024	Diagrama Pneumático Sistema de Paletização Robotizada	Quadro pneumático - QPU18	Lacticínios Tirol Ltda	+ Módulo geral
	Editor	Eduardo Augusto			18869 e 18871	Ipiranga - PR	Página 13 / 26
	Versão	3.0				

Vácuo Aplicador de papelão

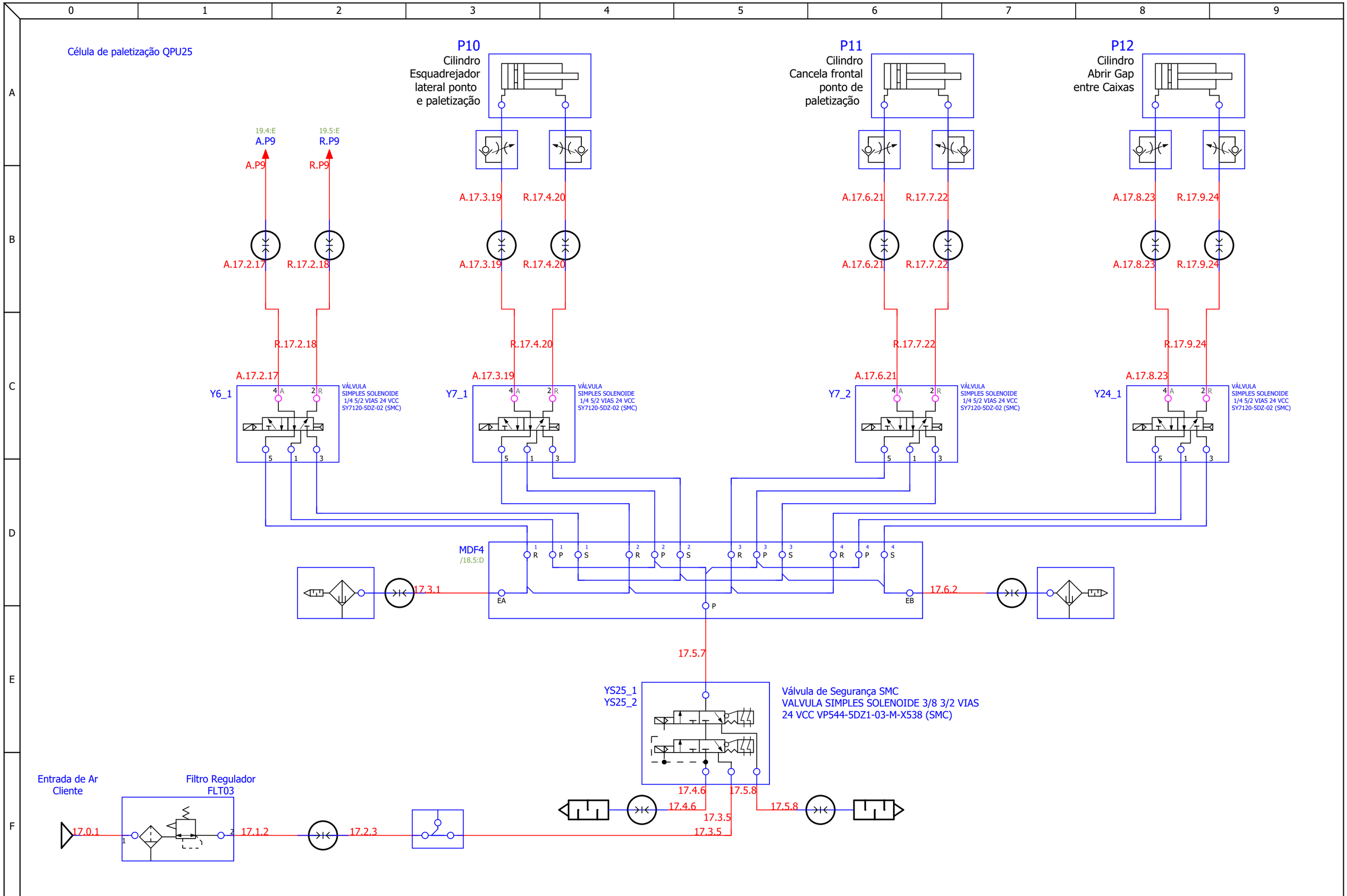


TORFRESMA	Data 28/10/2024	Diagrama Pneumático Sistema de Paletização Robotizada	Quadro pneumático - QPU18	Lacticínios Tirol Ltda	+ Módulo geral
	Editor Eduardo Augusto			18869 e 18871 / Ipiranga - PR	Página 14 / 26
	Versão 3.0				

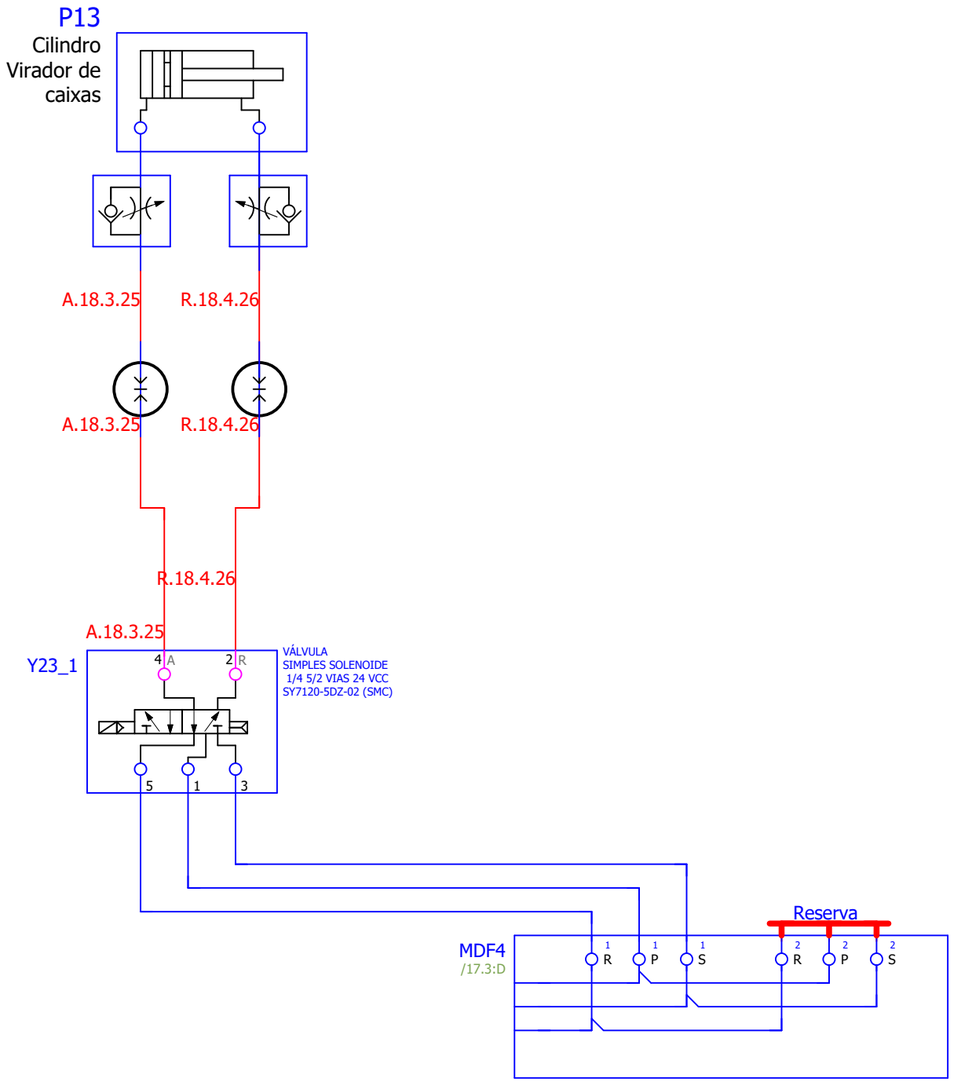


TORFRESMA	Data	28/10/2024	Diagrama Pneumático Sistema de Paletização Robotizada	Quadro pneumático - QPU18	Lacticínios Tirol Ltda	+ Módulo geral
	Editor	Eduardo Augusto			18869 e 18871 — Ipiranga - PR	Página 15 / 26
	Versão	3.0				

Reserva

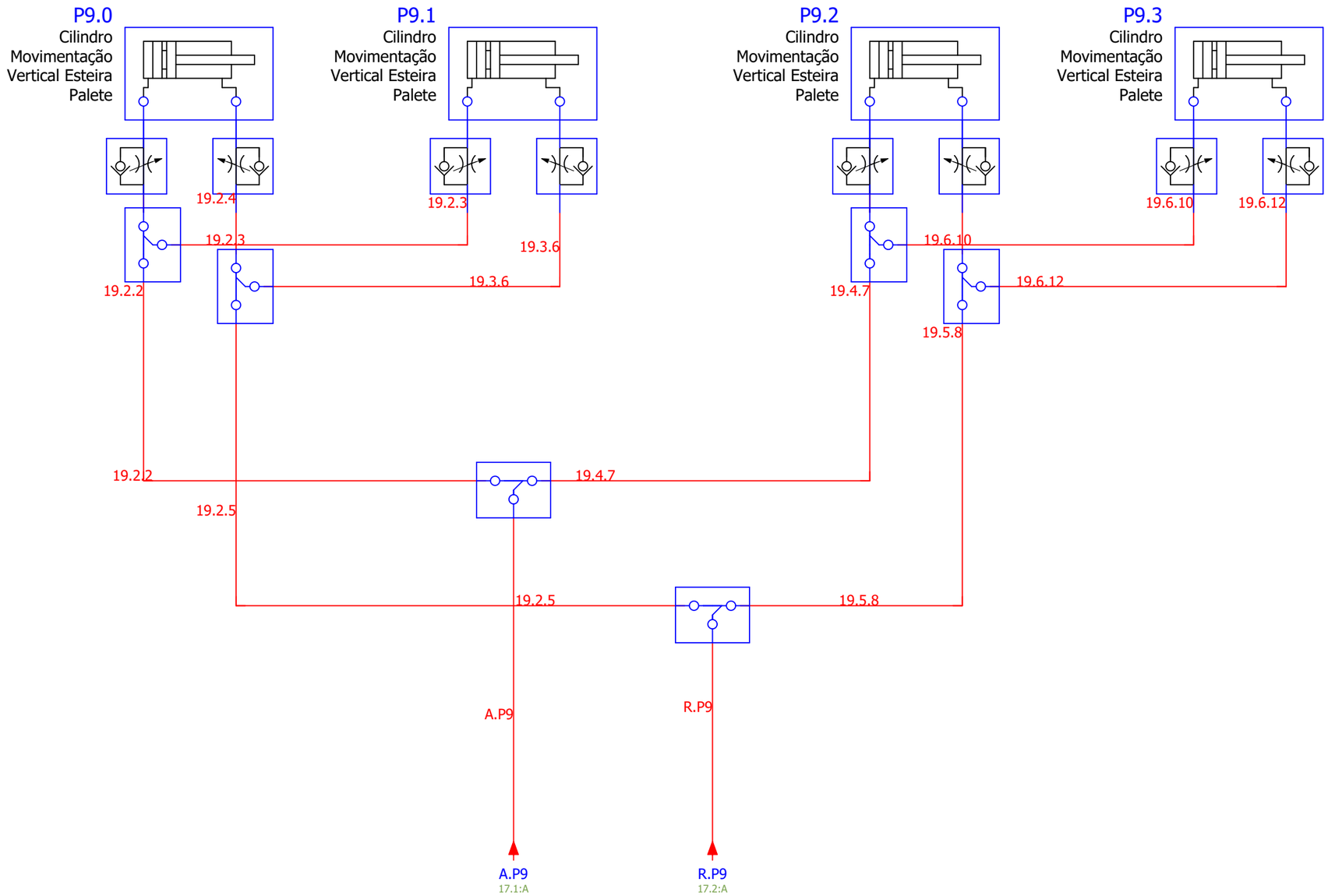


TORFRESMA	Data	28/10/2024	Diagrama Pneumático Sistema de Paletização Robotizada	Quadro pneumático - QPU25	Lacticínios Tirol Ltda	+ Módulo geral
	Editor	Eduardo Augusto			18869 e 18871	Ipiranga - PR	Página 17 / 26
	Versão	3.0				



TORFRESMA	Data	28/10/2024	Diagrama Pneumático Sistema de Paletização Robotizada	Quadro pneumático - QPU25	Lacticínios Tirol Ltda	+ Módulo geral
	Editor	Eduardo Augusto			18869 e 18871	Ipiranga - PR	Página 18 / 26
	Versão	3.0				

Esteira transportadora reta e captura

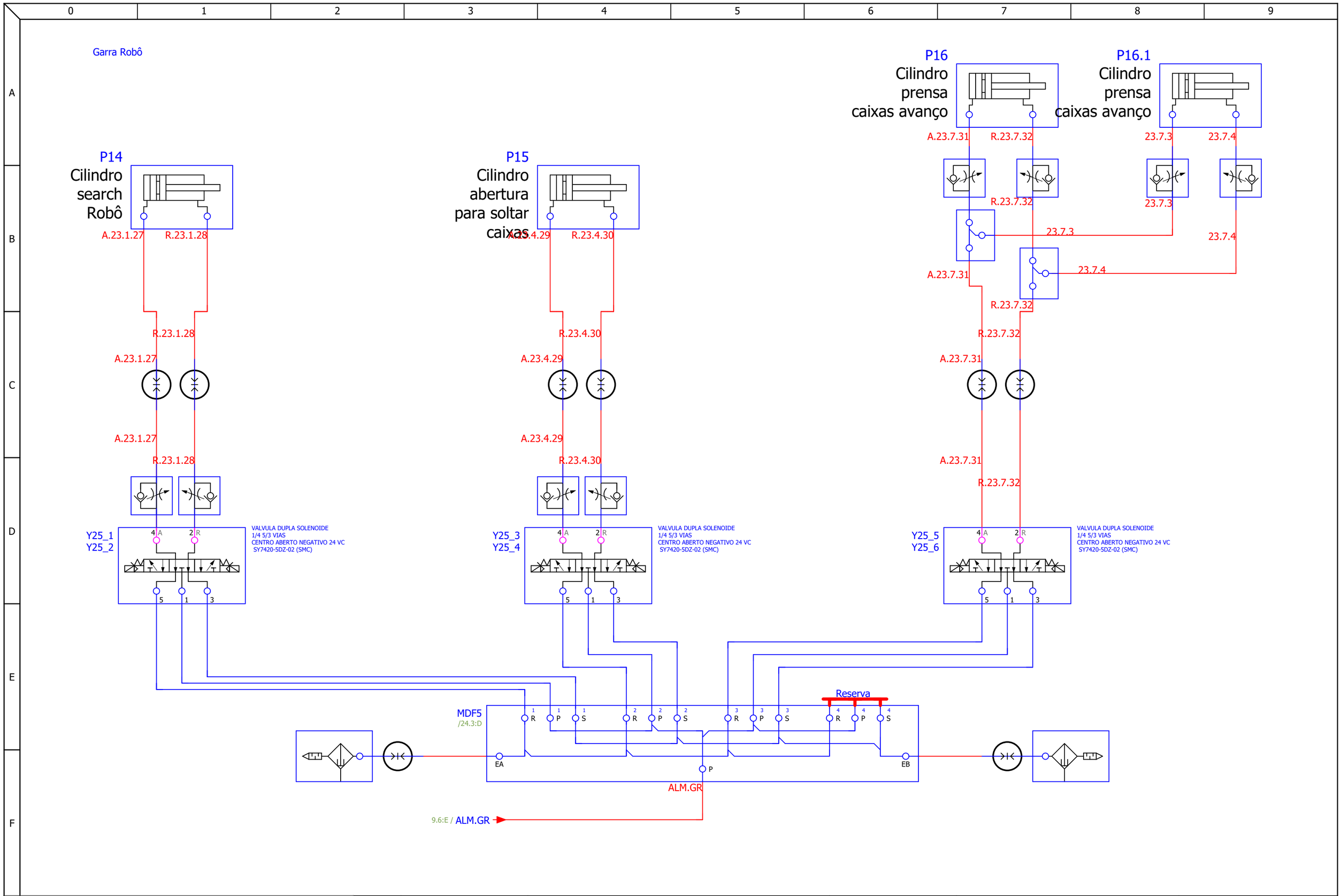


TORFRESMA	Data 28/10/2024	Diagrama Pneumático Sistema de Paletização Robotizada	Quadro pneumático - QPU25	Lacticínios Tirol Ltda	+ Módulo geral
	Editor Eduardo Augusto			18869 e 18871 / Ipiranga - PR	Página 19 / 26
	Versão 3.0				

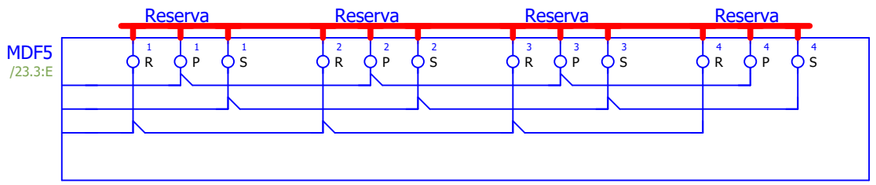
Reserva

Reserva

Reserva



TORFRESMA	Data	28/10/2024	Diagrama Pneumático Sistema de Paletização Robotizada	Garra Robô	Lacticínios Tirol Ltda	+ Módulo geral
	Editor	Eduardo Augusto			18869 e 18871	Ipiranga - PR	Página 23 / 26
	Versão	3.0			



TORFRESMA	Data	28/10/2024	Diagrama Pneumático Sistema de Paletização Robotizada	Garra Robô	Lacticínios Tirol Ltda	+ Módulo geral
	Editor	Eduardo Augusto			18869 e 18871	Ipiranga - PR	Página 24 / 26
	Versão	3.0			

Reserva

TORFRESMA	Data	28/10/2024	Diagrama Pneumático Sistema de Paletização Robotizada	Reserva	Lacticínios Tirol Ltda	+ Módulo geral
	Editor	Eduardo Augusto			18869 e 18871 / Ipiranga - PR	Página 25 / 26
	Versão	3.0				

Reserva

TORFRESMA	Data	28/10/2024	Diagrama Pneumático Sistema de Paletização Robotizada	Reserva	Lacticínios Tirol Ltda
	Editor	Eduardo Augusto			18869 e 18871 / Ipiranga - PR
	Versão	3.0			+ Módulo geral